swaps the x-axis and the z-axis and reverses the y-axis. One more distinction will be useful later. Multiplying an entire state by −1, or by any complex number of magnitude one, changes no measurement probability at all; such a global phase is physically invisible. Changing the sign of just one term in a superposition is different. That relative phase is precisely what distinguishes $|+\rangle$ from $|-\rangle$, and the Hadamard gate makes it measurable.

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} \qquad \begin{aligned} H|0\rangle &= |+\rangle, \quad H|1\rangle = |-\rangle \\ H|+\rangle &= |0\rangle, \quad H|-\rangle = |1\rangle \end{aligned}$$

$$H^2 = I, \qquad H = \frac{X+Z}{\sqrt{2}}, \qquad HXH = Z, \qquad HZH = X$$

$$H^{\otimes n}|0\cdots 0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle$$

Figure 3: The Hadamard matrix and some of its key identities

The real power of the Hadamard gate appears when it is applied to many qubits at once. Applying H to each of n qubits, all starting in $|0\rangle$, produces an equal superposition of all 2^n computational basis states, each with amplitude $1/\sqrt{(2^n)}$. With just ten gates, we can spread a ten-qubit register evenly across 1024 basis states. The matrix of this operation, written $H^{\otimes n}$, is older than the gate itself.

In 1867, James Joseph Sylvester showed how to build ever larger square arrays of +1s and −1s with mutually orthogonal rows: copy an array into the top-left, top-right and bottom-left quarters of a square array twice as wide and put its negative in the bottom-right quarter—the lonely minus sign of H, repeated on a larger scale. Starting from a single +1, the first doubling produces the pattern of H itself, and each further doubling adds one qubit; n doublings give the matrix of $H^{\otimes n}$, apart from the factor $1/\sqrt{(2^n)}$. Such arrays are now called Hadamard matrices, after Jacques Hadamard, who studied them in 1893, and the gate takes its name from them. In quantum computing, $H^{\otimes n}$ is the opening move of many well-known algorithms—and the source of a persistent myth, to which we shall return shortly.

Interference: Where amplitudes cancel

In the first decade of the 1800s, the English polymath Thomas Young showed that light passing through two narrow slits produces, on a screen behind them, a pattern of bright and dark bands. At the dark bands, light from one slit cancels light from the other: light added to light produces darkness. In 1961, Claus Jönsson repeated the experiment with electrons, and later versions sent the electrons through one at a time, only for the same bands to build up, dot by dot. Richard Feynman, whom we quoted in the first article, wrote that this experiment 'has in it the heart of quantum mechanics'.

We have already performed a version of this experiment. In the first article, we applied the Hadamard gate twice in the Bloch sphere simulator, watched the qubit return to $|0\rangle$, and took it as evidence that quantum gates are reversible. The first Hadamard gate turns $|0\rangle$ into $(|0\rangle + |1\rangle)/\sqrt{2}$. The second acts on each part separately: the $|0\rangle$ part becomes $(|0\rangle + |1\rangle)/2$ and the $|1\rangle$ part becomes $(|0\rangle - |1\rangle)/2$. There are now two routes to the outcome 0, contributing +1/2 and +1/2, which add up to 1. There are also two routes to the outcome 1, contributing +1/2 and −1/2, which add up to zero. The first is constructive interference, the second destructive. That is why the third article described the H–Z–H experiment as the essence of quantum interference: the Pauli-Z gate flips the sign of every route that passes through $|1\rangle$, so the cancellation moves to the outcome 0, and the qubit ends up in $|1\rangle$.

Here is the crux. Between the two Hadamard gates, the qubit is in a superposition of $|0\rangle$ and $|1\rangle$, and each part offers a route to the final outcome. As long as nothing measures the qubit in between, quantum mechanics tells us to add the amplitudes of the two routes first and to square only at the end. Probabilities are never negative, so combining them can only add up; amplitudes carry signs—and in general complex phases—so they can cancel. The TV show *MythBusters* once staged a memorable version of such cancellation: a soccer ball fired backwards at 60 mph from a truck travelling forwards at 60 mph simply dropped to the road. Velocities are not amplitudes, of course, but the lesson carries over: quantities with signs can add up to nothing. That single difference separates a quantum computer from an extremely expensive random number generator.

Interference is fragile, as the program *qiskit14.py*, shown below, reveals. It runs the H–H circuit twice: once undisturbed, and once with a sneaky measurement between the two gates.

```
1.  from qiskit import QuantumCircuit, transpile
2.  from qiskit_aer import AerSimulator
3.  sim = AerSimulator( )
4.  for peek in [False, True]:
5.      qc = QuantumCircuit(1, 2)
6.      qc.h(0)
7.      if peek:
8.          qc.measure(0, 0)
9.      qc.h(0)
10.     qc.measure(0, 1)
11.     compiled = transpile(qc, sim)
12.     counts = sim.run(compiled, shots=1000).result( )
`       .get_counts( )
13.     print("Peeking:" if peek else "No peeking:", counts)
```

Now let us understand how the program works. Line 4